

Metrics don't always measure what matters

Metric Design

Mahdi Nasser

Content

1. **What Metrics Do**
2. Metrics Design
3. Metrics Lifecycle
4. Right Metrics
5. Metrics Calculation
6. Metrics Interpretation
7. Measure what is not measurable

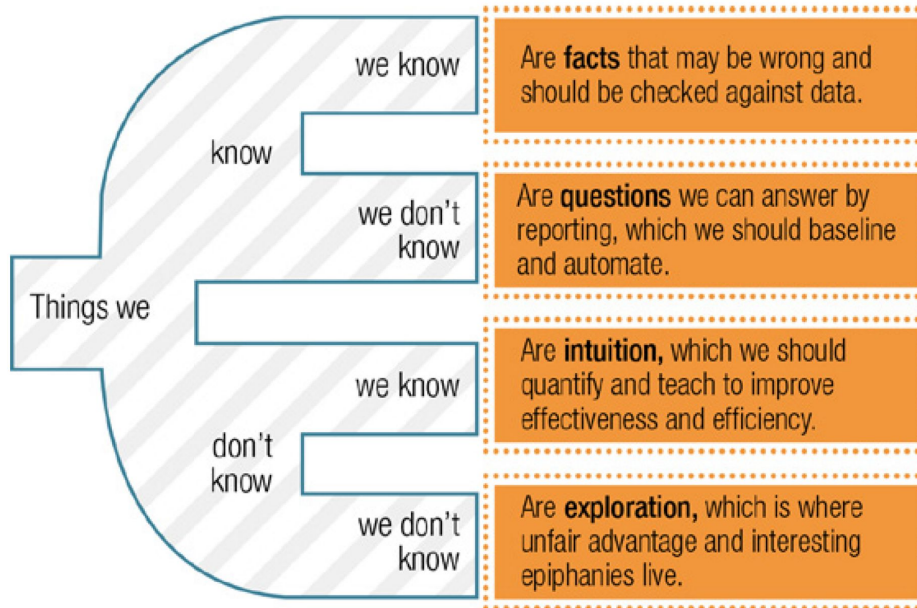
Uncertainty Does Matters

An “Information Theory” Version definition of Measurement:

Measurement: A quantitatively expressed reduction of uncertainty based on one or more observations.

It must reduce uncertainty and for that uncertainty reduction to be valuable **it must improve some decision.**

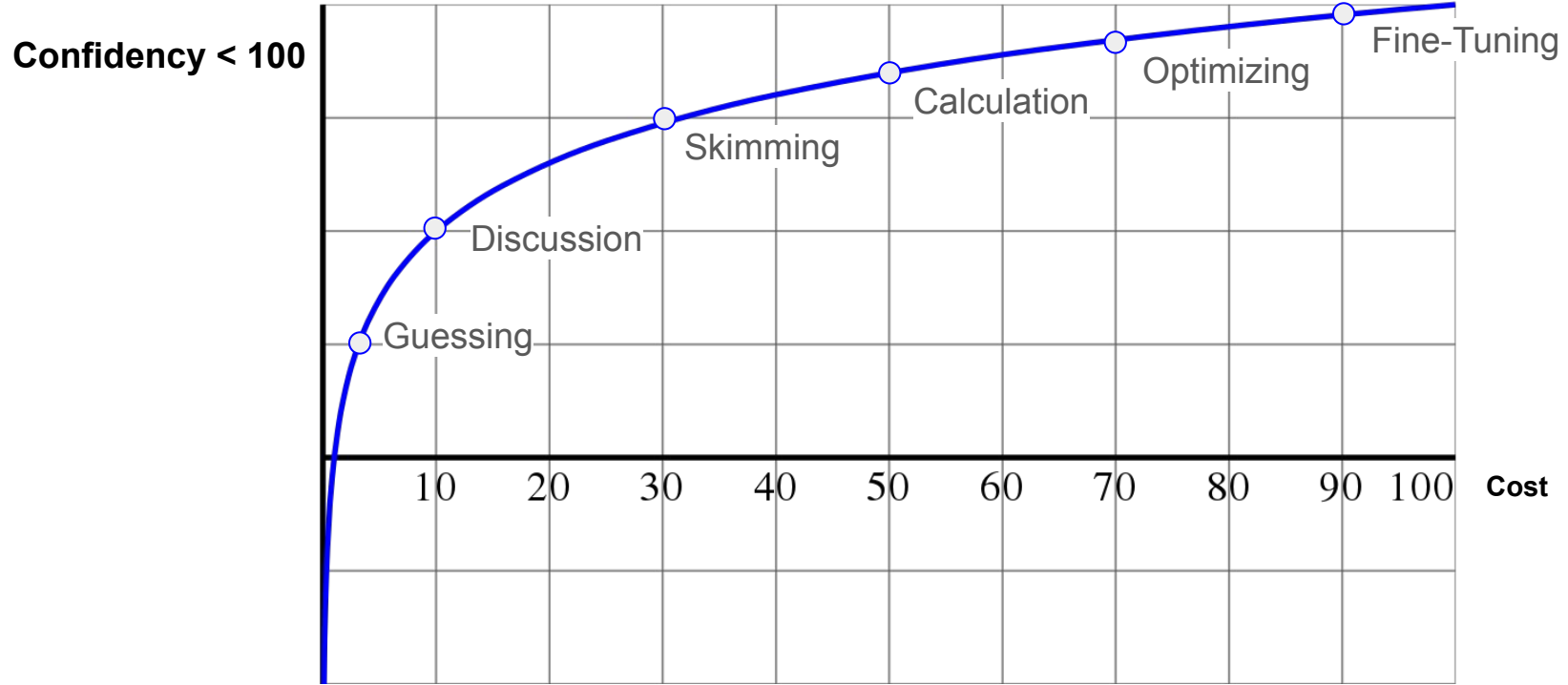
The hidden genius of Donald Rumsfeld



Four useful Measurement assumptions

1. It's been measured before.
2. You have far more data than you think.
3. You need far less data than you think.
4. Useful, new observations are more accessible than you think.

Confidence = LOG (Cost)



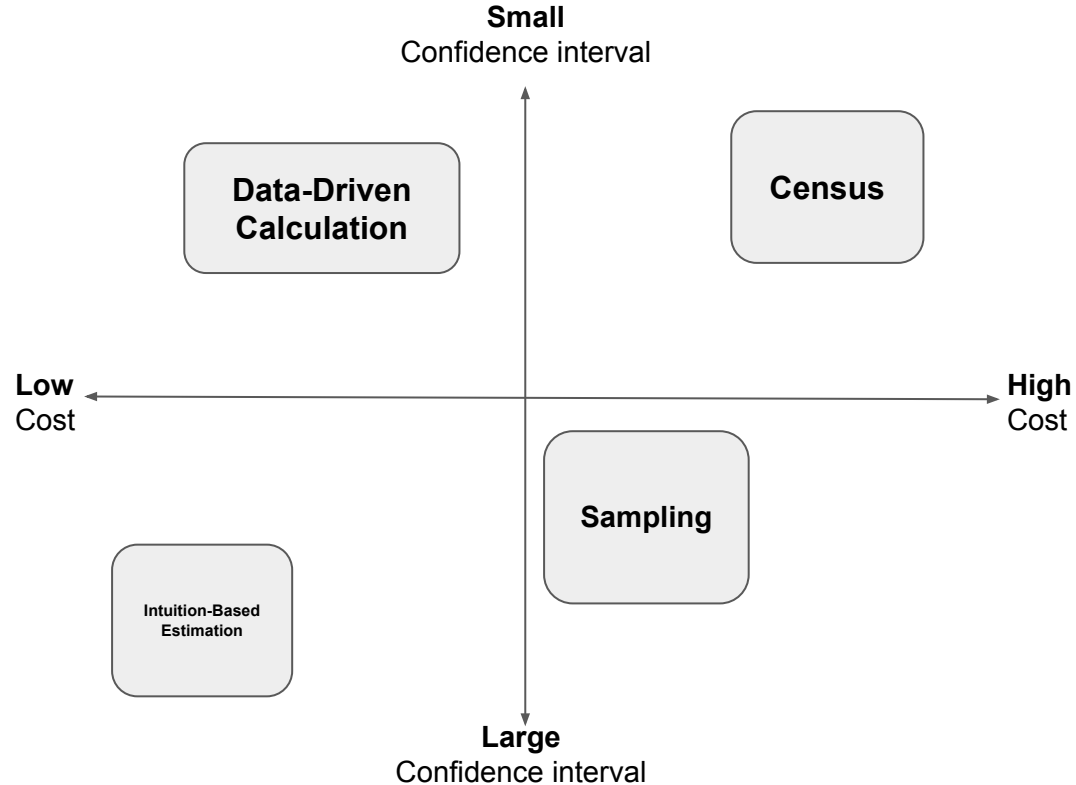
Sampling versus Census

Rule of Five: There is a **1/32** chance that the median of a population is between the smallest and largest values in any random sample of five from that population.

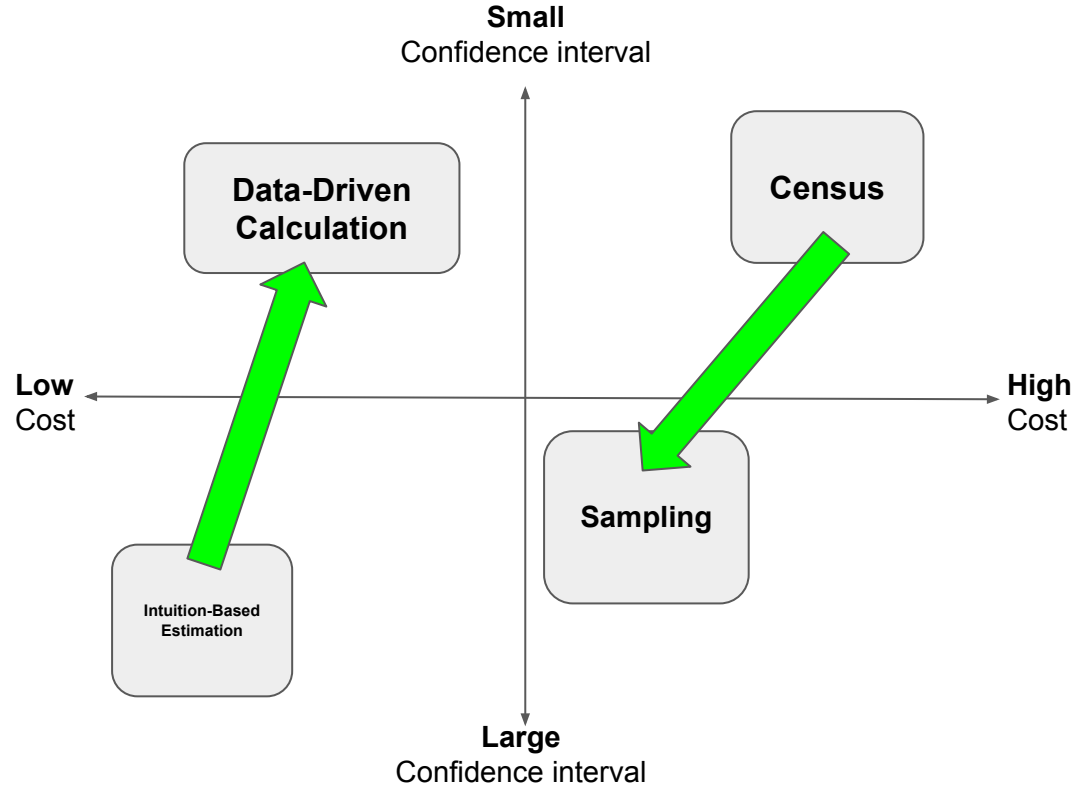
Sampling versus Census

Rule of Five: There is a 93.75% chance that the median of a population is between the smallest and largest values in any random sample of five from that population.

Measuring Strategies



Measuring Strategies



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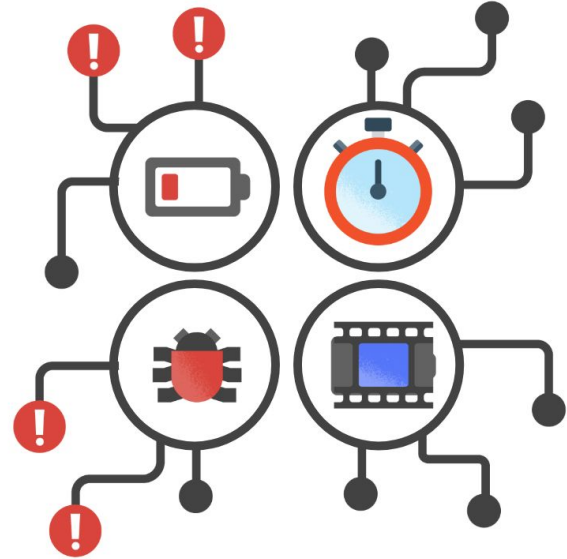
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Types of Metrics

Progress Metrics

Performance Metrics

Diagnostic Metrics



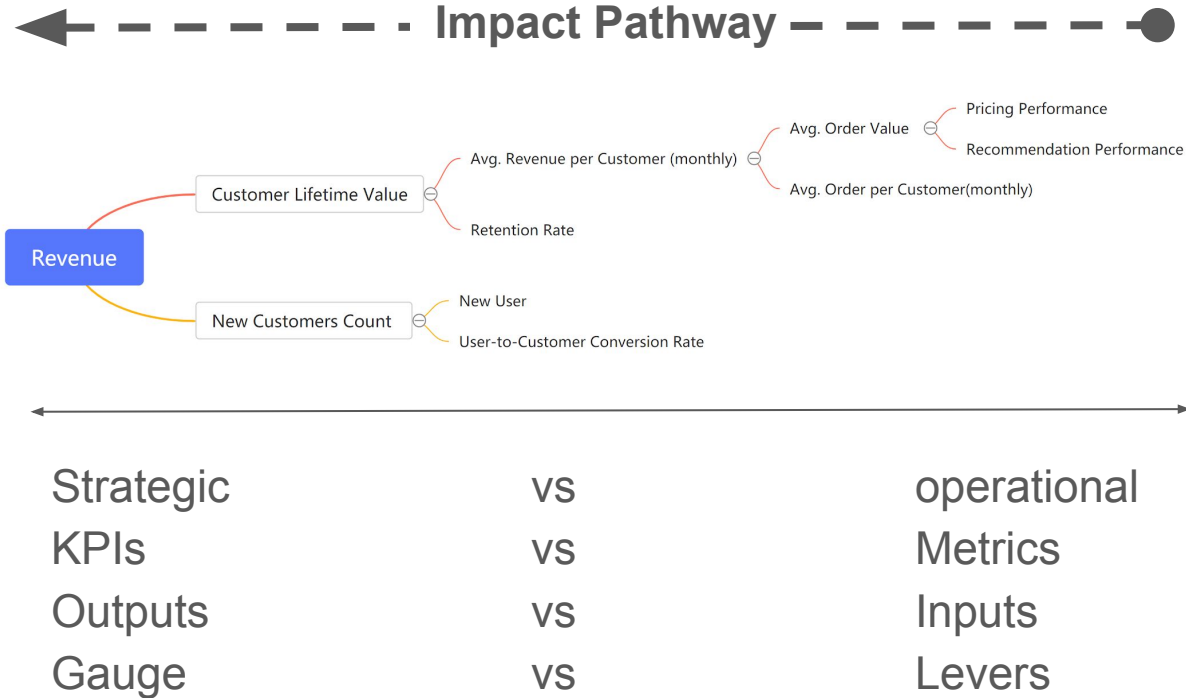
Metric Design Approaches

Impact Pathway

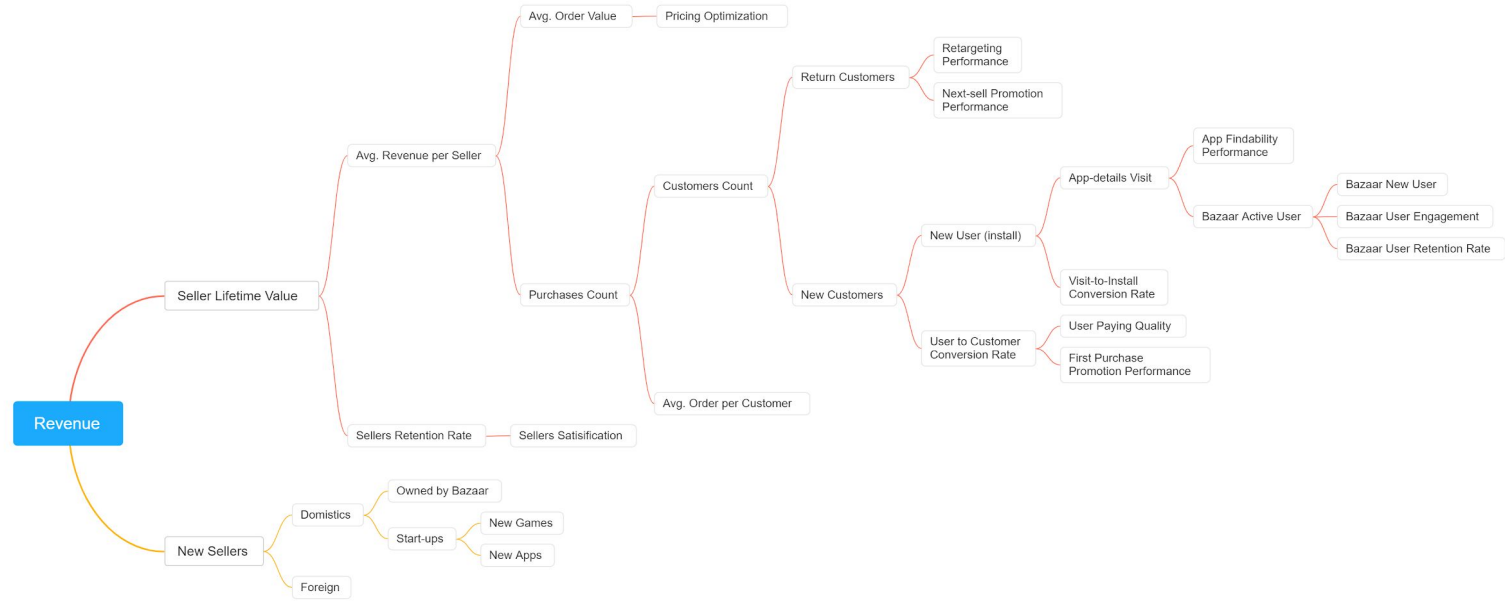
User Lifecycle

One Metric That Matters

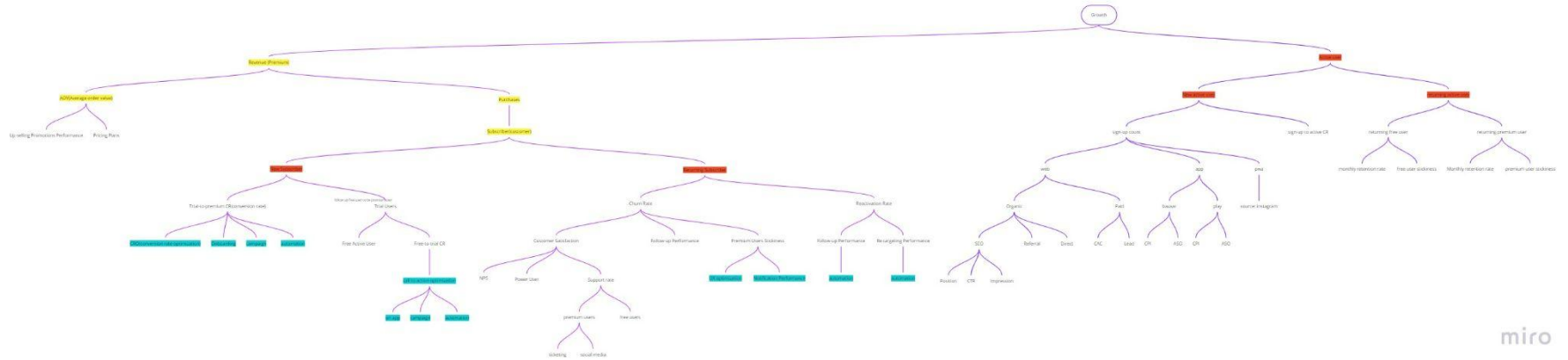
1. Impact Pathway



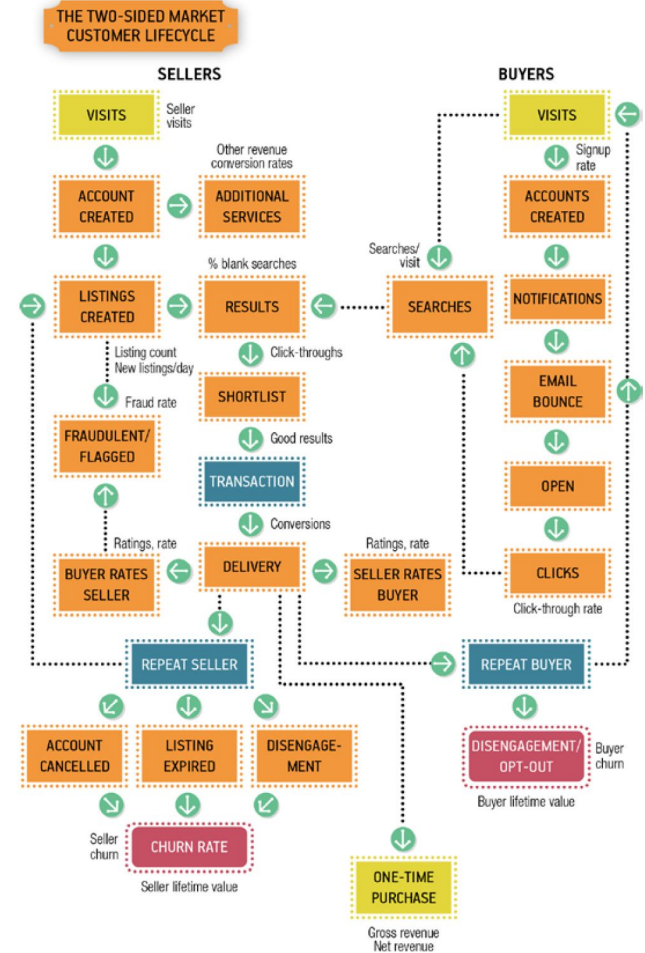
1. Impact Pathway



1. Impact Pathway



2. User lifecycle



3. One Metrics That Matters

مدل + مرحله = تنها سنجهای که برای رشد نیاز دارید



Metrics Cheat Sheet

Standard metrics Best Practices

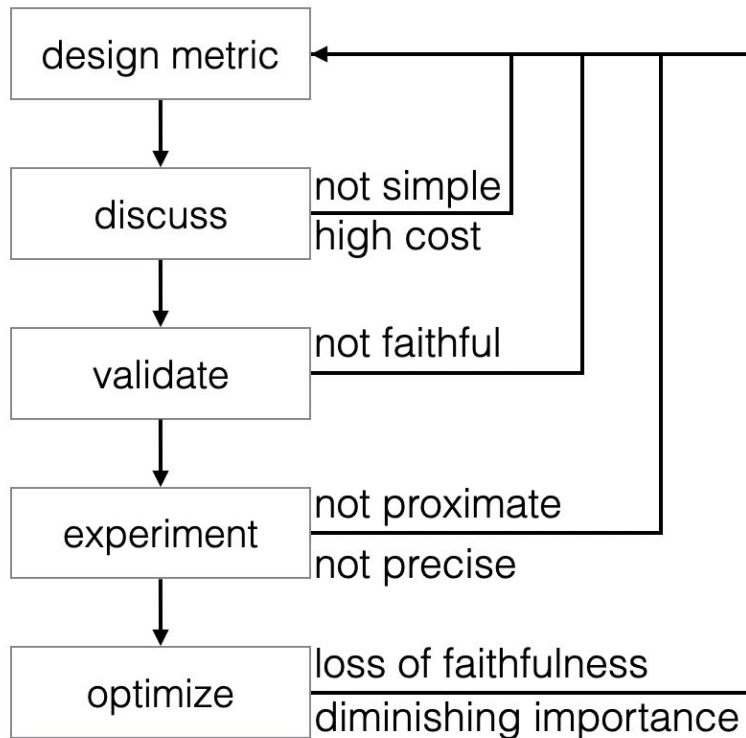
OR Make a new one if you believe on yourself

Financial performance	Understanding customers
• Net profit • Net profit margin • Gross profit margin • Operating profit margin • EBITDA • Revenue growth rate • Total shareholder return (TSR) • Economic value added (EVA) • Return on investment (ROI) • Return on capital employed (ROCE) • Return on assets (ROA) • Return on equity (ROE) • Debt-to-equity (D/E) ratio • Cash conversion cycle (CCC) • Working capital ratio • Operating expense ratio (OER) • CAPEX to sales ratio • Price earnings ratio (P/E ratio)	• Net Promoter Score (NPS) • Customer retention rate • Customer satisfaction index • Customer profitability score • Customer lifetime value (CLV) • Customer turnover rate • Customer engagement • Customer complaints

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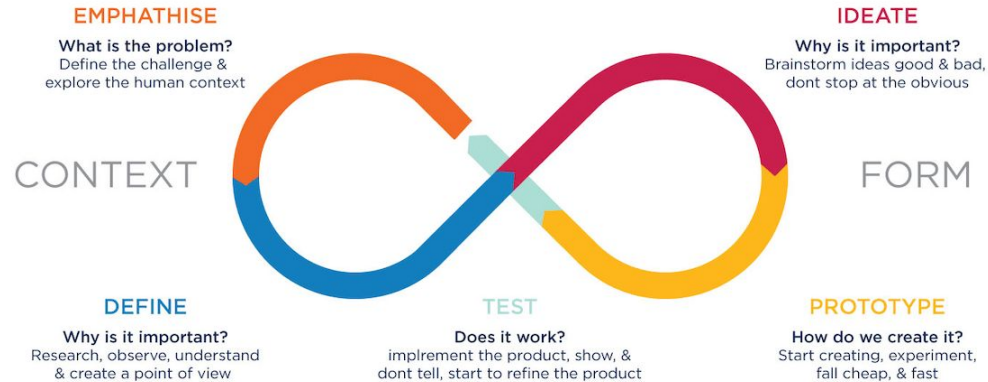
Metric Lifecycle



Think Like a Designer

DESIGN THINKING

A FRAMEWORK FOR INNOVATION



Metric Design Principles

Reusability

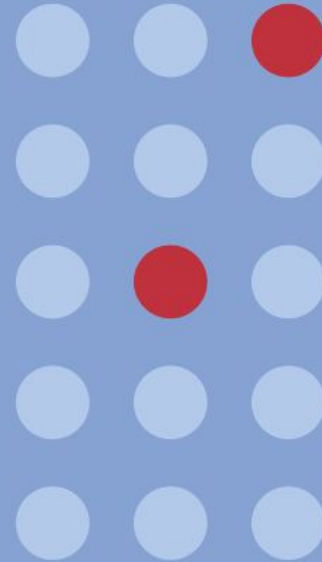
Repeatability

Interpretability

Effectivity

Simplicity

Comparability



Moving Targets

SMART!

draw a line in the sand, don't carve it in stone.

Use Benchmarks



Dashboards: Single Source of Truth

Centralized, Automated, Documented, Versioned, Shared

Dashboards need:

- Adjustment
- Development
- Maintenance

Avoid data vomit

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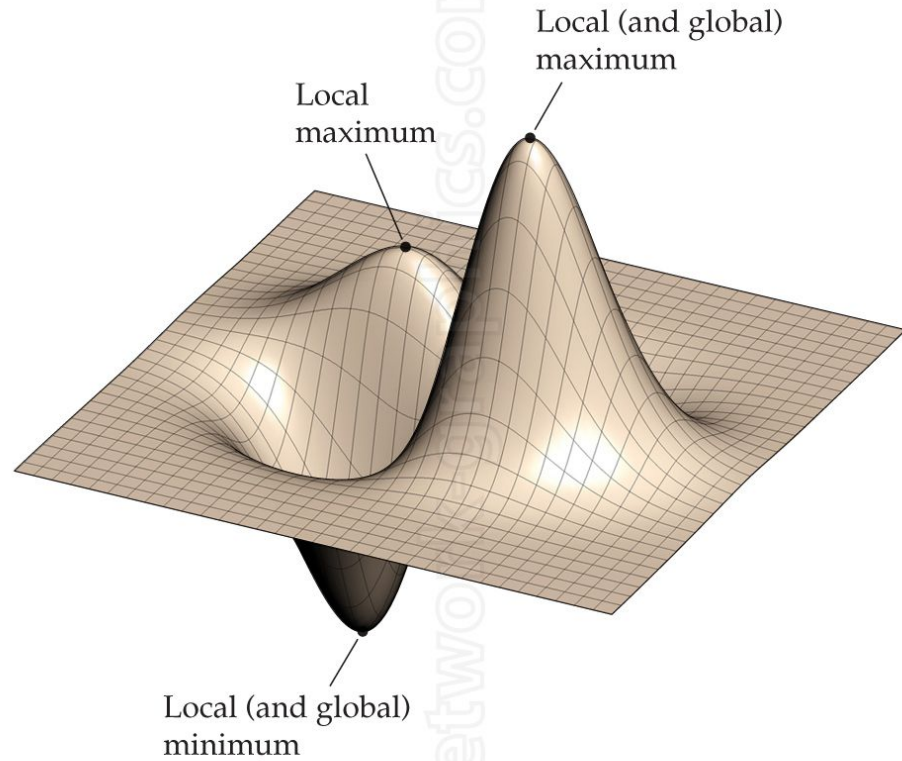
Golden Rule

A good metric changes the way you behave.

Don't measure what you can measure, measure what matters

If you can't cause a (statistically and practically) significant effect for your metric, then it is not very useful

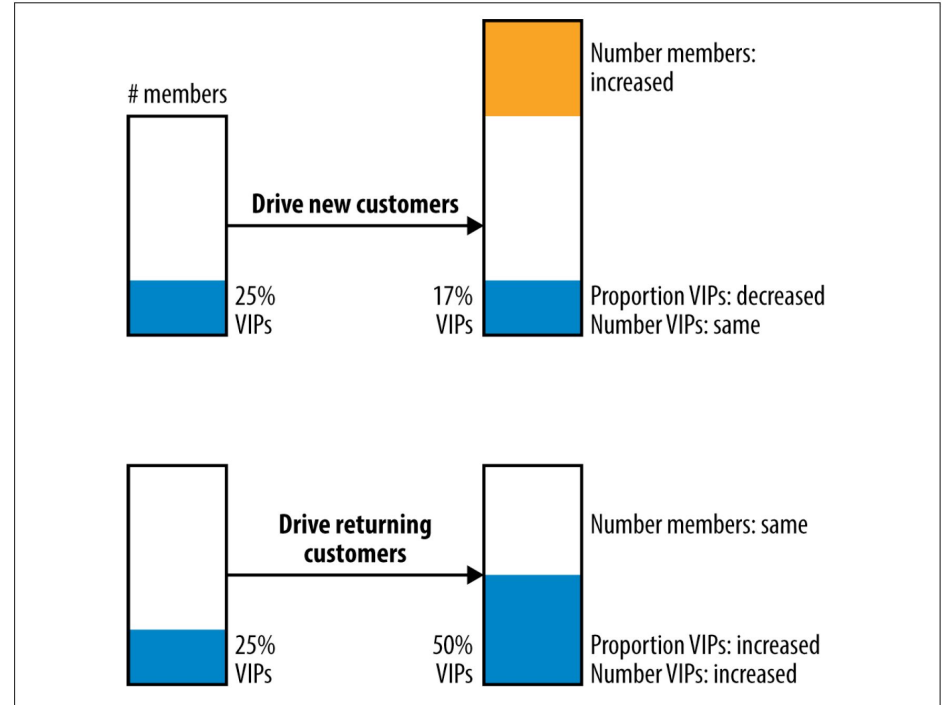
Local Optimization



Absolute versus Relative

Ratios are Actionable

Ratios are comparative



Growth and Size

Don't Ignore size when reporting growth

Context is critical.

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Data Cleansing

Cleaning the data you capture is often most of the work, and the simple act of cleaning it up can often reveal important patterns.

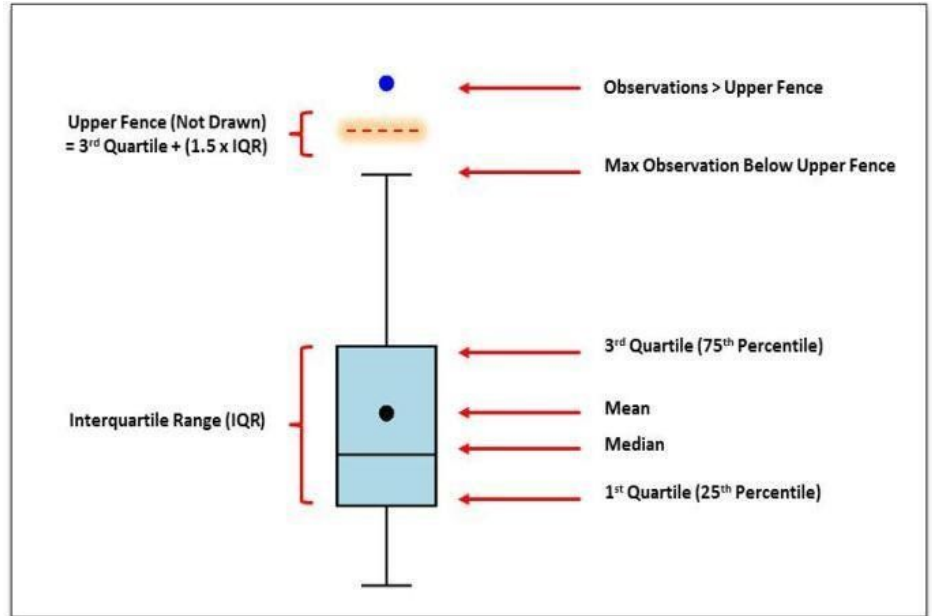
Calculation means Aggregation

Aggregation / summarizing

- Sum
- Count
- Average
- Ratio
- ...

Average is misleading

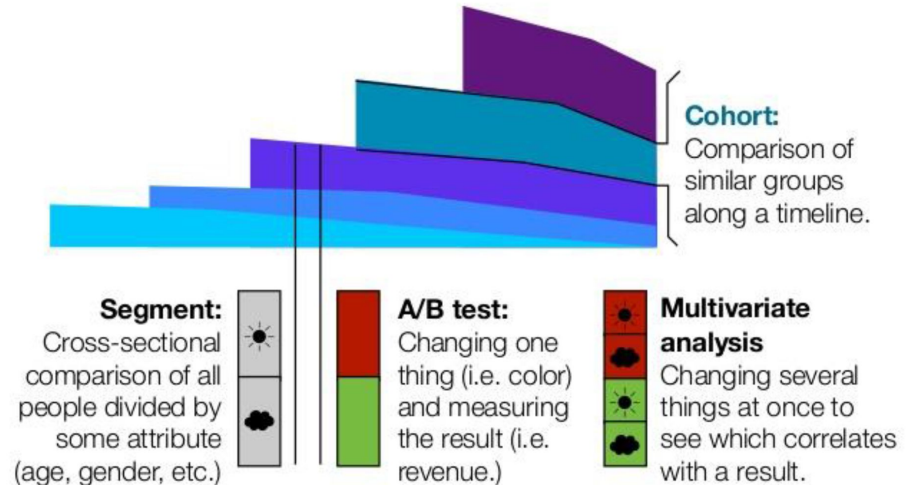
- Look at distributions
- Use Median or Quantiles
- OUTLIERS!



Accumulation is misleading

- Separate: Segments, Cohorts, Categories, etc.
- Normalize

Segments, cohorts, A/B, and multivariate



Outliers

Define metrics that are statistically “robust,” that is, they are relatively insensitive to individual extremal values.

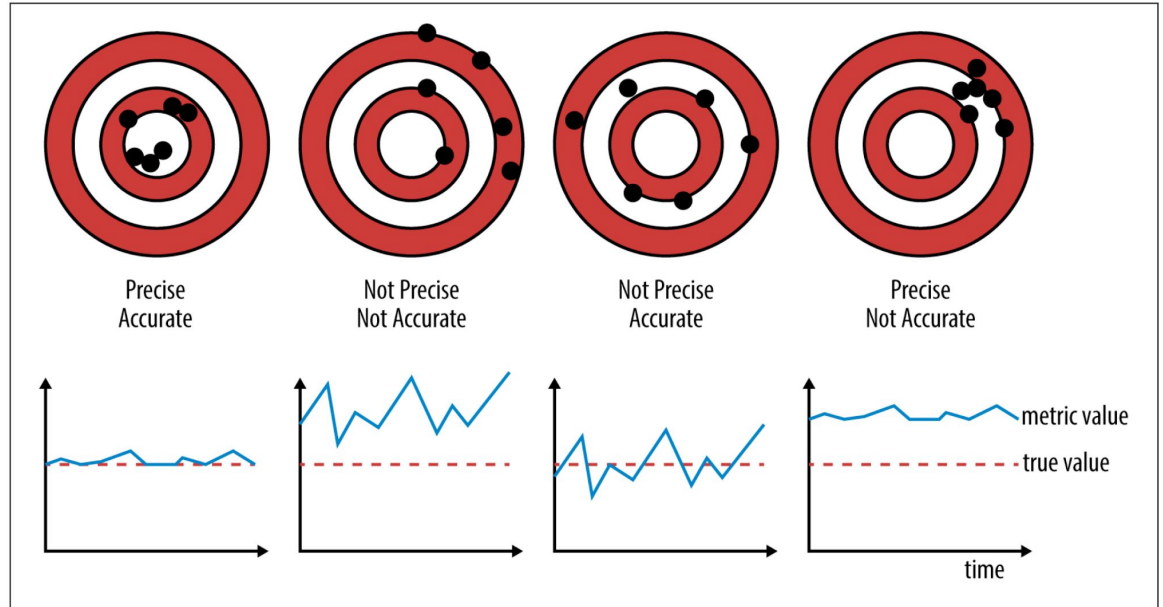
Outliers are interesting from a qualitative perspective, because they can show you things you didn't expect, they're not good for building a general model.

Use better aggregation: median instead of mean

Do EDA: histogram, boxplot

Remove Outliers (Possibly dangerous)

Robustness: Accuracy and precision



Normalize

Let's say you're making a list of popular wedding destinations. You could count the number of people flying in for a wedding, but unless you consider the total number of air travellers coming to that city as well, you'll just get a list of cities with busy airports.

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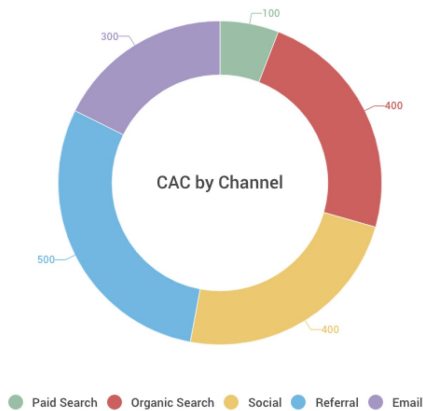
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Concern seasonality

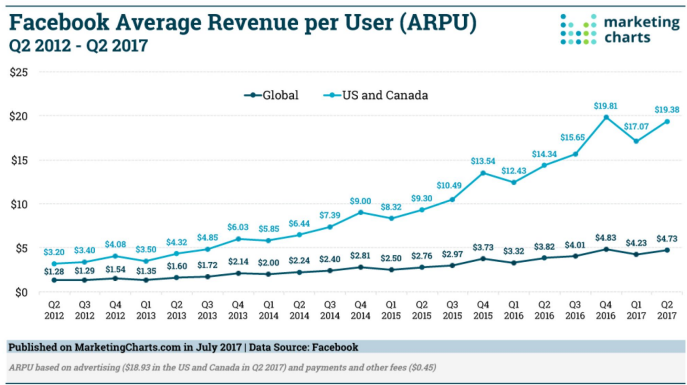
Failure to consider time of day, day of week, and monthly changes when looking at patterns leads to bad decision making.

Comparing

Compare Segments



Trace Trends



benchmarks

 Some interesting benchmarks

Growth
5% / week (revenue or active users)

Engaged visitors
30% monthly users
10% daily users

Time on site
17 minutes

Page load time
< 5 seconds

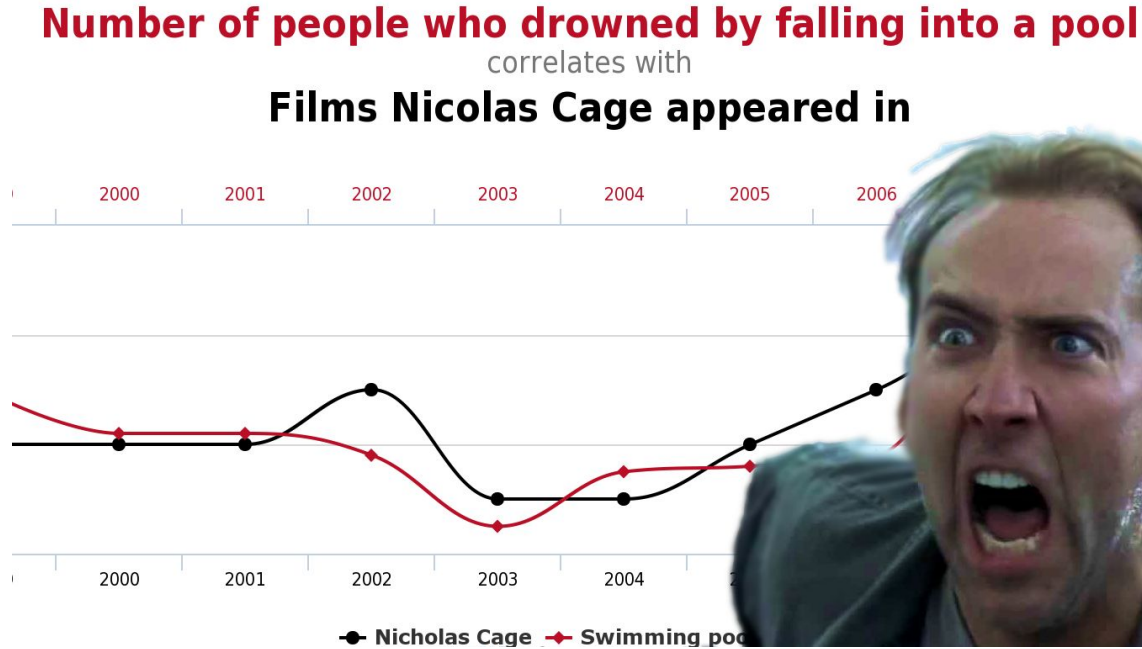
Free to paid
2% of free users

Churn
2% / month

Mobile file size
< 50MB

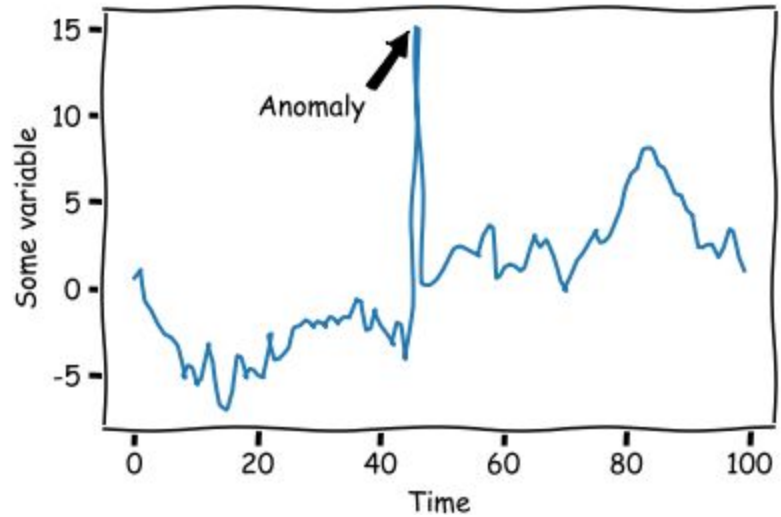
CLV:CAC
3:1

Causation vs correlation



Anomaly Detection

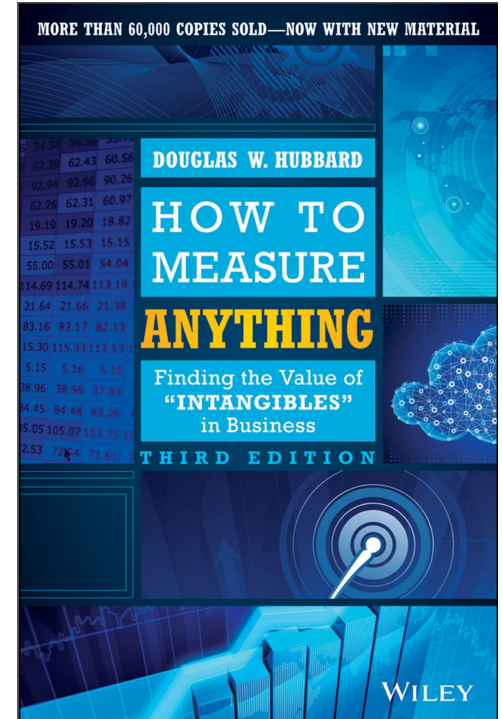
Set Alerts



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Measure what is not measurable



Direct versus Indirect

Proxy

Avoid proxies?!

Don't struggle in owned data

“Mashing up your data with data from other sources can lead to valuable insights